SOURCE CODE

```
int pirPin = 2;     // PIR sensor output pin

int ledPin = 13;    // LED pin

int motionState = 0;


void setup() {

  pinMode(pirPin, INPUT);   // PIR as input

  pinMode(ledPin, OUTPUT);  // LED as output

  Serial.begin(9600);       // Start serial monitor

}


void loop() {

  motionState = digitalRead(pirPin);  // Read sensor


  if (motionState == HIGH) {

    digitalWrite(ledPin, HIGH);  // LED ON

    Serial.println("Motion Detected!");

  }

  else {

    digitalWrite(ledPin, LOW);   // LED OFF

    Serial.println("No Motion");

  }


  delay(500);  // Small delay

}
```

EXPLANATION

A PIR (Passive Infrared) sensor detects motion by sensing changes in infrared radiation emitted by human bodies. Normally, it monitors the surroundings and stays LOW. When a person moves in front of it, the heat pattern changes, and the sensor output becomes HIGH. This change is used to detect motion.

In the Arduino code, the PIR sensor is connected to a digital input pin and continuously checked using `digitalRead()`. When the sensor output is HIGH (motion detected), the Arduino turns ON the LED and prints “Motion Detected!” on the Serial Monitor. If the output is LOW (no motion), the LED remains OFF and it prints “No Motion”. A small delay is added to stabilize readings.

Overall, the system reads sensor data, displays the result through LED and SerialMonitor, and triggers an action (LED ON) when motion crosses the threshold.

CIRCUIT DIAGRAM